

SHAILESH PADALKAR

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SUMMARY

Mechanical Engineering Ph.D. candidate (expected Dec 2027) specializing in real-time 3D and virtual reality training systems for manufacturing. Most of my work centers on building immersive VR training in Unreal Engine, from modeling equipment in Siemens NX through to interaction design and deployment on Meta Quest Pro, in collaboration with partners such as Intel, Siemens, and GE Aerospace. I developed a set of VR cleanroom training modules now used to train students and educators in Ohio's semiconductor workforce program, and I recently published first-author research on integrating large language models into VR manufacturing training. I also work with CAD-to-scan registration and inspection pipelines.

TECHNICAL SKILLS

Extended Reality (XR) & Real-Time 3D: Unreal Engine 5 (Blueprint visual scripting - advanced; C++ - working knowledge), VR interaction design (grab/release, teleportation, widget interaction, event-driven object behaviors), 3D graphics (meshes, PBR materials and material instances, textures, metallic/roughness/emissive workflows, Lumen lighting), immersive environment setup, Meta Quest Pro deployment, eye-tracking integration

3D Graphics & CAD Asset Pipelines: Siemens NX (3D modeling, assemblies), NX Open API (C++), CAD-to-real-time asset workflows (NX to OBJ to Unreal Engine), 3D scene assembly, geometric feature recognition, faceted (STL) to parametric CAD reconstruction, 1:1 facility-scale environment modeling.

AI / LLM Integration: LLM-driven application pipelines (OpenAI GPT-4o-mini, Google Gemini 2.5 Flash), retrieval-augmented generation (RAG), sentence-transformer embeddings (all-MiniLM-L6-v2), JSON-schema-constrained generation, Whisper automatic speech recognition (ASR), manager-worker LLM orchestration, prompt engineering.

Inspection and Metrology: Point cloud registration (Iterative Closest Point), KD-tree nearest-neighbor search, correspondence filtering, CAD-scan alignment, PMI extraction, GD&T tolerance evaluation.

Programming: Unreal Engine Blueprint, C++ (Unreal Engine, NX Open API), Python, MATLAB, C# (Unity)

Software and Tools: Unreal Engine 5, Unity, Siemens NX, Meta Quest Pro, Visual Studio, Dassault Systemes 3DEXPERIENCE

EDUCATION

Ph.D., Mechanical Engineering, University of Cincinnati, Cincinnati, OH	Aug 2021 - Dec 2027 (expected)
Advisor: Prof. Sam Anand Research focus: real-time 3D visualization, VR training systems, LLM integration for manufacturing, and digital twins	
M.Tech., Mechanical Engineering, University of Mumbai, India	2011 - 2013
B.E., Mechanical Engineering, University of Mumbai, India	2006 - 2010

RESEARCH EXPERIENCE

Graduate Research Assistant, Smart Manufacturing Lab, University of Cincinnati	May 2022 - Present
Intel OASiS Semiconductor Cleanroom VR Training (Partner: Intel Corporation) / Dec 2022 - Present	
- Lead end-to-end development of real-time VR training applications within a \$4M Intel-funded OASiS semiconductor workforce development program, spanning CAD modeling, 3D asset preparation, interaction development, testing, and deployment. - Designed, developed, integrated, tested, and deployed 7 immersive VR training modules covering semiconductor chip manufacturing and cleanroom operations using Unreal Engine Blueprint and supporting C++ classes. - Established a repeatable CAD-to-real-time 3D asset pipeline by modeling cleanroom equipment and a full-scale facility in Siemens NX and converting, importing, and assembling the assets in Unreal Engine via OBJ. - Engineered custom VR interaction systems beyond the native Unreal VR template, including object-specific grabbing, 3D widget interactions, and event-driven machine behaviors triggered by user actions. - Developed reusable PBR master materials and material instances for real-time 3D environments, incorporating base-color and imperfection textures with parameterized metallic, roughness, and emissive properties. - Integrated multimodal video and audio feedback systems to guide users through cleanroom procedures and improve training clarity.	

- Delivered hands-on VR training to 60+ high-school students and 9 educators, supporting a multi-institute semiconductor workforce development initiative featured by the University of Cincinnati.

LLM-Assisted VR Training for Manufacturing (Partners: Siemens Digital Industries, Cincinnati Incorporated, Valvoline) / Jan 2025 - Dec 2025

- Architected a modular, LLM-driven framework that converts unstructured standard operating procedures (SOPs) and spoken user queries into structured, executable VR training steps; published as first author at ASME IDETC-CIE 2026.
- Built a manager-worker LLM orchestration layer (GPT-4o-mini, Gemini 2.5 Flash) with Whisper-based speech recognition that routes queries across three intent classes (procedure generation, general Q&A, widget interaction).
- Implemented a RAG pipeline (all-MiniLM-L6-v2 embeddings, 600-token chunks, top 3 cosine-similarity retrieval) with JSON-schema-constrained output and provenance metadata to prevent hallucinated instructions in safety-critical training.
- Integrated Meta Quest Pro eye tracking for component identification and a sphere-of-influence spatial filter with a custom Unreal Blueprint class to ground LLM instructions to the correct virtual machine components in cluttered scenes.
- Developed a deterministic, state-based runtime monitoring engine with action-specific verification (distance tracking, gaze dwell, control-state flags, positional error, and snap-to-place with 5 cm tolerance) for stable step-by-step progression.
- Validated the framework on three industrial case studies (press brake gage alignment, atomic layer deposition, semiconductor wafer resistivity measurement), achieving up to 0.92 F1-score in procedural step alignment against expert reference SOPs; pre-generated execution plans reduced runtime LLM calls, latency, and token cost.

IIoT-Based Augmented Reality for Legacy Manufacturing / 2022–2023

- Developed an augmented reality application in Unity using C# to visualize real-time manufacturing machine data within an interactive AR environment.
- Integrated IIoT machine data with the Unity-based AR interface, enabling users to access process information and data-driven insights directly within the augmented environment.
- Contributed to the development and validation of the AR/IIoT framework, resulting in a peer-reviewed publication in Manufacturing Letters, 2023.

As-Built Digital Twin for Automated Inspection (Partners: GE Aerospace, Siemens, Cincinnati Incorporated) / Jan 2026 - Present

- Own the CAD-to-scan registration stage of a multi-partner inspection pipeline that aligns 3D scans of manufactured parts to nominal CAD, maps PMI data onto the aligned geometry, and evaluates GD&T tolerance conformance.
- Developed an Iterative Closest Point (ICP) registration workflow in Python and NX Open API C++, incorporating KD-tree nearest-neighbor search and a custom active-deactivate correspondence filtering pipeline to improve alignment robustness.
- Support downstream PMI extraction and mapping to registered geometry, enabling automated pass/fail tolerance evaluation of as-built parts.

CAD Feature Recognition and Reconstruction (Research contract: Raytheon Technologies) / May 2022 - Dec 2022

- Co-developed (2-person team) computational tools in Siemens NX and NX Open API C++ to convert faceted STL geometry into parametric native CAD (.PRT) models for reuse of legacy mesh data in production workflows.
- Responsible for identifying geometric features in triangulated STL data and reconstructing the corresponding parametric CAD features while preserving geometric fidelity.

TEACHING & PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant, University of Cincinnati, Cincinnati, OH

Aug 2021 - May 2022

- Assisted in three graduate/undergraduate courses across the year: Manufacturing Processes (Fall 2021), Control Systems (Spring 2022), and a hands-on lab course (Summer 2022).
- Supported course delivery through grading, student assistance, and leading laboratory sessions.

Assistant Professor, Vishwakarma Institute of Technology (VIT), Pune, India

Jun 2015 - May 2020

- Taught undergraduate courses in Mechatronics and Finite Element Methods, and mentored students both academically and personally.
- Served on the development team for a departmental laboratory and contributed to related administrative responsibilities.

- Guided student projects that secured competitive research grants from Dassault Systèmes totaling INR 180,000.

Contractual Faculty, Veermata Jijabai Technological Institute (VJTI), Mumbai, India

Jun 2013 - May 2015

- Taught undergraduate courses in Control Systems and Manufacturing Planning & Control.
- Conducted hands-on laboratory instruction in Pro/ENGINEER (CAD), 3D printing, and Engineering Drawing.

ADDITIONAL PROJECTS

- **VR Children's Play Area (course project):** Designed and built an interactive VR play-area environment in Unreal Engine, including scene layout, materials, and Blueprint-scripted object interactions.
- **VR Classroom Environment (course project):** Created an immersive classroom scene in Unreal Engine with interactive objects to explore VR-based learning-space design.

SELECTED PUBLICATIONS

- **S. Padalkar**, S. Polisetty, A. Gautam, M. Rezayat, S. Anand, "A Modular, LLM-Driven Framework for Scalable Virtual Reality Manufacturing Training," ASME IDETC-CIE 2026 (accepted; camera-ready submitted).
- A. Gautam, M. R. Aryal, S. Deshpande, **S. Padalkar**, M. Nikolaenko, M. Tang, S. Anand, et al., "IIoT-Enabled Digital Twin for Legacy and Smart Factory Machines with LLM Integration," Journal of Manufacturing Systems, vol. 80, pp. 511-523, 2025.
- S. Deshpande, **S. Padalkar**, S. Anand, "IIoT-Based Framework for Data Communication and Prediction Using Augmented Reality for Legacy Machine Artifacts," Manufacturing Letters, vol. 35, pp. 1043-1051, 2023.

AWARDS, CERTIFICATIONS, AND RECOGNITION

- NSF Student Travel Award, North American Manufacturing Research Conference (NAMRC), 2023
- University of Cincinnati Graduate Incentive Award, 2021 - Present
- Dassault Systemes 3DEXPERIENCE Certified Associate
- Research featured in University of Cincinnati news: "UC begins training students, workers in microchip manufacturing" (July 2024)